

Name: _____ Student #: _____

Q1: Trace the following program using the chart method you learned about in your workshop

1) Create the charts and trace the program, updating variables in the chart (2.5 marks)

2) Determine the exact output of the program (2.5 marks)

```
int main(void)
{
    int apple = 10;
    int orange = 20;
    printf("A = %d\n", apple++);
    printf("B = %d\n", --orange);
    printf("C = %d\n", apple);
    printf("D = %d\n", orange);
    printf("E = %d\n", 3 + 2 * 4 % 3);
}
```

Q2: Write a program that will ask the user for a measurement in inches . You may assume that the measurement is to the nearest whole inches. Your program will output the measurement in centimeters. (5 marks)

Note: 1 inch = 2.54 cm. For the prompt use: "Enter the number of inches". For output, be sure the use the format "X" is Y cm where X is the number the user entered and Y is calculated result

Sample run (this is just an example, your program must work for any input):

```
Enter the number of inches: 5
5" is 12.7 cm
```